Discrete Foundations of Emergent Spacetime and Gravity

Universal Grid Updates, Quantum Correspondence, and the Finite-Response Principle

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Abstract

Grid-Clock Cosmology (GCC) is formulated as a discrete-substrate framework in which a fixed graph of elementary adjacency relations and a universal integer-valued update index are more fundamental than physical distance and physical time. Proper time, spatial length, causal propagation, quantum evolution, and gravitational geometry are treated as coarse-grained observables of the grid state. In a reference vacuum, one update is calibrated to the Planck time and one elementary link to the Planck length, while the local null speed satisfies

$$c = \frac{\ell_{\mathrm{P}}}{t_{\mathrm{P}}}.$$

The microscopic ordering is preferred, but local Lorentz symmetry is required to emerge in the low-energy, long-wavelength limit. Closed-system quantum evolution is represented by local unitary updates,

$$|\Psi_{K+1}\rangle = U_K|\Psi_K\rangle, \qquad U_K^{\dagger}U_K = I.$$

Spatially extended coherent states are denoted by the effective V sector, while localized, metastable, record-bearing states are denoted by the effective M sector. This distinction is an effective organization of grid states rather than a derivation of the Standard Model. Gravity is represented by the dependence of the emergent lapse, shift, and spatial metric on the coarse-grained processing state. The weak-field limit is required to reproduce universal geodesic motion and the Einstein--Hilbert correspondence.

A minimum cosmological sector is retained only as a bridge to the companion observational paper. Its equation of state,

$$w_G(a) = -\frac{1}{1 + (a_t/a)^n},$$

has an exact spatially flat ΛCDM correspondence at $n = 3$. The constant baseline grid count N_{grid} is distinguished from a possible effective active count $N_{\mathrm{eff}}(K, \mathbf{N})$; variation of the latter is classified as an open extension relevant to directional dependence or cyclic evolution, not as a result established in this paper. The manuscript separates foundational postulates, correspondence results, effective ansatzes, and open predictions, and states explicit falsification and no-retuning requirements.

Keywords: *emergent spacetime; discrete time; quantum cellular dynamics; quantum gravity; effective dark sector; finite response; falsifiability.*

1 Foundational Postulates and Kinematical Structure

1.1 Scope and methodological status

Grid-Clock Cosmology is proposed as an effective foundational framework connecting discrete microscopic evolution to relativistic spacetime, quantum dynamics, gravity, and cosmology. Its purpose is not to derive every particle species or every interaction of the Standard Model. The present construction addresses six questions:

1. How can physical time and physical distance emerge from a more elementary discrete system?
2. How can local Lorentz kinematics arise when the microscopic theory possesses an ordered sequence of updates?
3. How can unitary quantum evolution be represented on the same substrate?
4. How can gravitational geometry be interpreted as a response of the substrate rather than as an additional force inside a pre-existing spacetime?
5. How can a fixed number of elementary grid units produce an expanding cosmological geometry?
6. Which observations can distinguish the construction from general relativity plus ΛCDM?

The theory separates three descriptive levels:

$$\text{microscopic grid and clock} \rightarrow \text{emergent geometry and quantum fields} \rightarrow \text{macroscopic observables.} \quad (1.1)$$

Physical coordinates, proper time, spatial length, particle trajectories, and cosmological redshift belong to the emergent or macroscopic levels. They are not assumed to be fundamental properties of the microscopic substrate.

1.2 Fundamental discrete substrate

1.2.1 Definition 1: Elementary grid

The microscopic spatial substrate is represented by a graph

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}), \tag{1.2}$$

where $\mathcal{V}$ is the set of elementary grid units and $\mathcal{E}$ is the set of allowed adjacency relations. The number of elementary grid units is

$$N_{grid} = |\mathcal{V}|. \tag{1.3}$$

The first foundational postulate is

$$\boxed{N_{grid} = \text{constant.}} \tag{1.4}$$

Cosmic expansion, gravitational deformation, and quantum evolution do not require the continuous creation or destruction of elementary grid units. They are represented by changes in the physical state associated with the vertices and edges of $\mathcal{G}$.

The framework does not initially require a choice between a very large finite graph and a countably infinite graph. A finite graph is convenient when discussing conservation of total processing capacity and cyclic cosmological evolution. Local continuum results need not depend on this global distinction.

1.2.2 Postulate 1: Relabeling invariance

Labels assigned to individual vertices and edges have no direct physical meaning. Observable quantities must be invariant under graph relabelings that preserve adjacency and dynamical structure. This is the discrete analogue of coordinate redundancy.

1.3 Universal update index

1.3.1 Definition 2: Fundamental clock

The microscopic state evolves through an integer-valued update index

$$K \in \mathbb{Z}. \tag{1.5}$$

The complete microscopic state at update K is denoted by $\mathcal{X}_K$, and its evolution is represented abstractly as

$$\mathcal{X}_{K+1} = \mathcal{F}_K[\mathcal{X}_K]. \tag{1.6}$$

The ordering $K < K + 1$ is fundamental, whereas the physical duration associated with one update is not. Different regions may assign different proper-time intervals to the same change in K, depending on their local grid state.

1.3.2 Postulate 2: Monotonic microscopic ordering

All physical histories admit a consistent ordering by increasing K. The macroscopic arrow of time is not identified with this ordering alone; irreversibility additionally requires the formation of records and the dispersal of correlations into inaccessible environmental degrees of freedom.

1.4 Locality and finite propagation

1.4.1 Postulate 3: Microscopic locality

An elementary update at a vertex may depend directly only on that vertex and a finite neighborhood defined by graph adjacency. For a nearest-neighbor theory,

$$\mathcal{X}_i(K + 1) = \mathcal{F}_i\big[\mathcal{X}_i(K), \{\mathcal{X}_j(K)\}_{j\sim i}\big]. \tag{1.7}$$

Microscopic locality produces a finite causal propagation speed. In the reference vacuum, the maximum propagation of an elementary disturbance is one grid link per update.

1.5 Reference-vacuum calibration

The reference vacuum is not an absolute state of nonexistence. It is defined operationally as a homogeneous ground configuration with vanishing *additional* processing load. In this state,

$$\Delta\tau_{ref} = t_P, \qquad \Delta\ell_{ref} = \ell_P. \tag{1.8}$$

A null disturbance advances by one elementary link in one update,

$$\Delta N = 1, \qquad \Delta K = 1, \tag{1.9}$$

so that

$$\boxed{c = \frac{\ell_P}{t_P}.} \tag{1.10}$$

This is a calibration condition. Massive modes and interacting excitations may propagate more slowly, while the local causal speed must not exceed the null limit in the low-energy effective theory.

The Planck relation

$$E_P t_P = \hbar \tag{1.11}$$

motivates interpreting $\hbar$ as the reference action scale associated with one elementary cell and one elementary clock interval. This does not derive the measured numerical value of $\hbar$; it identifies its role in translating dimensionless microscopic phase into physical units.

1.6 Emergent spacetime variables

The effective spacetime geometry is obtained by coarse-graining the microscopic grid state. Three quantities are required:

$$\alpha(K, \mathbf{N}), \qquad \beta^a(K, \mathbf{N}), \qquad h_{ab}(K, \mathbf{N}). \tag{1.12}$$

Here α is the emergent lapse or local clock-rate field, β^a is the effective shift or flow of the spatial grid state, and h_{ab} is the dimensionless spatial metric associated with grid separations.

The effective line element is postulated to be

$$\boxed{ds^2 = -\alpha^2 c^2 t_P^2\, dK^2 + \ell_P^2 h_{ab}(dN^a + \beta^a dK)(dN^b + \beta^b dK).} \tag{1.13}$$

This equation is a kinematical map, not yet a dynamical law. The dynamics determining α, β^a, and h_{ab} are introduced through the grid-state field and its effective action.

For a worldline comoving with the effective spatial flow,

$$dN^a + \beta^a dK = 0, \tag{1.14}$$

and therefore

$$\boxed{d\tau = \alpha t_P\, dK.} \tag{1.15}$$

For neighboring points on a constant-K hypersurface,

$$\boxed{d\ell^2 = \ell_P^2 h_{ab}\, dN^a dN^b.} \tag{1.16}$$

Thus, the number of microscopic clock updates and grid links remains unchanged, while their physical interpretation depends on the state of the emergent geometry.

1.7 Homogeneous and isotropic limit

For a homogeneous and isotropic background with vanishing shift,

$$\beta^a = 0, \qquad h_{ab} = a^2(K) b^2(\chi) \gamma_{ab}, \tag{1.17}$$

where $a(K)$ is the cosmological scale factor, $b(\chi)$ is a local spatial-response function, χ is a coarse-grained grid-state variable, and γ_{ab} is a maximally symmetric reference metric.

For a radial displacement,

$$d\ell = a(K) b(\chi) \ell_P\, dN, \tag{1.18}$$

and

$$d\tau = \alpha(\chi)t_P\, dK. \tag{1.19}$$

Cosmic expansion is therefore defined by

$$\frac{d}{dK}[a(K)\ell_P] \neq 0, \tag{1.20}$$

rather than by $dN_{grid}/dK \neq 0$.

1.8 Local invariance of the speed of light

For radial null propagation in the isotropic limit,

$$0 = -\alpha^2 c^2 t_P^2 dK^2 + a^2 b^2 \ell_P^2 dN^2. \tag{1.21}$$

Using $c = \ell_P/t_P$ gives the microscopic coordinate velocity

$$\boxed{\frac{dN}{dK} = \frac{\alpha}{ab}.} \tag{1.22}$$

The locally measured physical speed is

$$\frac{d\ell}{d\tau} = \frac{ab\ell_P\, dN}{\alpha t_P\, dK} = \frac{\ell_P}{t_P} = c. \tag{1.23}$$

Thus, the coordinate speed on the grid may vary while the locally measured speed remains invariant. The local clock, ruler, and null trajectory are governed by the same emergent geometry.

1.9 Flat-grid relativistic limit

In the flat reference state,

$$\alpha = 1, \qquad \beta^a = 0, \qquad h_{ab} = \delta_{ab}. \tag{1.24}$$

For one spatial direction define

$$\boxed{u = \frac{v}{c} = \frac{\Delta N}{\Delta K}.} \tag{1.25}$$

The proper clock count satisfies

$$\boxed{(\Delta K_\tau)^2 = (\Delta K)^2 - (\Delta N)^2.} \tag{1.26}$$

Therefore,

$$\boxed{\gamma = \frac{\Delta K}{\Delta K_\tau} = \frac{1}{\sqrt{1-u^2}}.} \tag{1.27}$$

This relation is the discrete starting point for the relativistic derivations in Chapter 2.

1.10 Discrete quantum state

The quantum state on the grid at update K is

$$|\Psi_K\rangle = \textstyle\sum_N \psi_N(K)|N\rangle. \tag{1.28}$$

For a closed system,

$$\boxed{|\Psi_{K+1}\rangle = U_K|\Psi_K\rangle,} \tag{1.29}$$

with

$$\boxed{U_K^\dagger U_K = I.} \tag{1.30}$$

If

$$U_K = \exp\left(-\frac{it_P}{\hbar}\hat{H}_K\right), \tag{1.31}$$

the long-wavelength continuum limit yields the Schrödinger equation. In a nontrivial grid state, the conversion between K and proper time is controlled by the lapse.

1.11 Effective V and M Sectors

The V sector contains states characterized by spatial extension, coherent phase relations, propagation, delocalized superposition, and dynamically active correlations. The M sector contains relatively localized, stable or metastable, record-bearing states.

The sectors are not assumed to be two unrelated substances. They are effective phases or sectors of the microscopic state. A conversion

$$V \leftrightarrow M \tag{1.32}$$

represents a change in localization, coherence, and stability.

The total energy is decomposed schematically as

$$\boxed{E_V + E_M + E_{grid} + E_{int} = E_{tot}.} \tag{1.33}$$

For a closed system,

$$\Delta_K\left(E_V + E_M + E_{grid} + E_{int}\right) = 0. \tag{1.34}$$

1.12 Decoherence and localization

For two alternative histories with action difference ΔS, the relative phase is

$$\Delta\phi = \frac{\Delta S}{\hbar}. \tag{1.35}$$

For approximately Gaussian environmental fluctuations, the coherence factor is

$$|\Gamma| = \exp\left[-\frac{Var(\Delta S)}{2\hbar^2}\right]. \tag{1.36}$$

Define

$$\boxed{\mathcal{D} = \frac{Var(\Delta S)}{2\hbar^2}.} \tag{1.37}$$

Then $|\Gamma| = e^{-\mathcal{D}}$. The regime $\mathcal{D} \ll 1$ preserves coherent V-like behavior, whereas $\mathcal{D} \gtrsim 1$ suppresses interference. Decoherence alone does not derive the selection of one outcome or the Born rule; an effective M state also requires localization, dynamical stability, and persistent record formation.

1.13 Inertia and the arrow of time

A freely propagating particle is represented as a stable moving pattern of microscopic updates. If q_K denotes its coarse-grained configuration and $u_K = q_{K+1} - q_K$, inertial motion satisfies

$$\Delta_K u_K = 0. \tag{1.38}$$

Inertia is therefore the persistence of a stable translation pattern. Inertial mass measures resistance of the pattern to changes in propagation phase, direction, or internal update structure.

The ordering of K does not alone explain thermodynamic irreversibility. The effective arrow of time arises from

$$\text{ordered updates} + \text{record formation} + \text{environmental dispersion of correlations}. \tag{1.39}$$

Entropy is interpreted operationally as microscopic information inaccessible under a specified coarse-graining.

1.14 Isotropy as a background solution

The fundamental grid need not be exactly isotropic at every scale. Write

$$\alpha = \bar{\alpha}(1 + \delta_\alpha), \qquad \beta^a = \delta\beta^a, \tag{1.40}$$

and

$$h_{ab} = a^2(K)[\gamma_{ab} + q_{ab}]. \tag{1.41}$$

The exactly homogeneous and isotropic limit is

$$\delta_\alpha = 0, \qquad \delta\beta^a = 0, \qquad q_{ab} = 0. \tag{1.42}$$

A residual vector field may generate dipolar effects, while a trace-free tensor component may generate quadrupolar or higher-order angular dependence. The distinction is

$$\text{anisotropy permitted by the framework} \neq \text{anisotropy required by a derived solution.} \tag{1.43}$$

A minimum observational parameterization is

$$H(z,\hat{\mathbf{n}}) = H_{iso}(z)\exp[A_H g_H(z)\hat{\mathbf{n}}\cdot\hat{\mathbf{p}}]. \tag{1.44}$$

For $|A_H| \ll 1$,

$$H(z,\hat{\mathbf{n}}) \simeq H_{iso}(z)[1 + A_H g_H(z)\hat{\mathbf{n}}\cdot\hat{\mathbf{p}}]. \tag{1.45}$$

1.15 Minimum consistency conditions

A viable Grid-Clock Cosmology must satisfy:

- low-energy Lorentz correspondence;
- ordinary quantum correspondence;
- universal gravitational response;
- Newtonian and relativistic gravitational limits;
- a stable reference vacuum;
- cosmological viability at both background and perturbation levels;
- parameter economy and a no-retuning rule;
- a directional null test for any derived anisotropic solution;
- an explicit turnaround and bounce solution before cyclic evolution is claimed.

The next chapter derives relativistic kinematics from the discrete interval and establishes the conditions under which Lorentz transformations emerge.

2 Grid-Clock Relativity

2.1 Purpose and assumptions

The microscopic Grid-Clock framework contains an elementary graph $\mathcal{G}$ and a universal integer-valued update index K. This structure differs conceptually from ordinary special relativity, in which no preferred microscopic update ordering is assumed. The existence of a fundamental update index does not, however, imply that an observable preferred frame must appear at accessible energies. The central requirement is that ordinary Lorentz symmetry emerge as an effective symmetry in the low-energy and long-wavelength regime.

The derivation uses four effective assumptions:

1. **Homogeneous reference grid:** $\alpha = 1$, $\beta^a = 0$, and $h_{ab} = \delta_{ab}$.
2. **Isotropic null propagation:** at wavelengths much larger than the grid spacing, the maximum signal speed is the same in all directions.
3. **Linearity of inertial transformations:** homogeneous inertial transformations are linear in the coarse-grained variables.
4. **Emergent relativity principle:** stable low-energy grid patterns describe local nongravitational physics using the same effective laws.

These are not asserted to be exact symmetries of individual microscopic vertices. They are requirements on the coarse-grained observables.

2.2 Dimensionless grid-clock interval

For one spatial dimension let ΔK be the number of microscopic update intervals between two events and ΔN their signed grid separation. The flat reference-grid interval is

$$\boxed{\Delta\Sigma^2 = -(\Delta K)^2 + (\Delta N)^2.} \tag{2.1}$$

For a timelike trajectory define the proper clock count by

$$\boxed{(\Delta K_\tau)^2 = (\Delta K)^2 - (\Delta N)^2.} \tag{2.2}$$

The physical quantities are

$$\Delta t = t_P \Delta K, \qquad \Delta x = \ell_P \Delta N, \qquad \Delta\tau = t_P \Delta K_\tau. \tag{2.3}$$

Using $c = \ell_P / t_P$,

$$\boxed{\Delta s^2 = -c^2 \Delta t^2 + \Delta x^2 = -c^2 \Delta\tau^2.} \tag{2.4}$$

The ordinary Minkowski interval is therefore the physical-unit representation of the dimensionless grid-clock interval. This relation is an effective causal interval obtained after coarse-graining many microscopic updates; it is not a Euclidean path length on the graph.

2.3 Null propagation and invariant speed

For a null trajectory,

$$\Delta s^2 = 0, \tag{2.5}$$

and therefore

$$(\Delta K)^2 = (\Delta N)^2. \tag{2.6}$$

For propagation in the positive direction,

$$\frac{\Delta N}{\Delta K} = 1. \tag{2.7}$$

The physical velocity is

$$v_{null} = \frac{\Delta x}{\Delta t} = \frac{\ell_P \Delta N}{t_P \Delta K} = c. \tag{2.8}$$

A massive grid pattern satisfies $|\Delta N| < |\Delta K|$, while a null pattern satisfies equality. A rule with $|\Delta N| > |\Delta K|$ lies outside the effective causal cone.

2.4 Dimensionless velocity and Lorentz factor

For a stable moving pattern define

$$\boxed{u = \frac{\Delta N}{\Delta K} = \frac{v}{c}.} \tag{2.9}$$

Substituting $\Delta N = u\Delta K$ into the proper-clock relation gives

$$\Delta K_\tau = \Delta K\sqrt{1-u^2}. \tag{2.10}$$

The Lorentz factor is therefore

$$\boxed{\gamma(u) = \frac{\Delta K}{\Delta K_\tau} = \frac{1}{\sqrt{1-u^2}}.} \tag{2.11}$$

In physical variables,

$$\boxed{\gamma(v) = \frac{1}{\sqrt{1-v^2/c^2}}.} \tag{2.12}$$

2.5 Emergent Lorentz transformation

Consider inertial systems S and S', with the origin of S' moving at dimensionless velocity u relative to S. The most general linear transformation in one spatial dimension is

$$K' = AK + BN, \qquad N' = CK + DN. \tag{2.13}$$

The trajectory of the primed origin is $N = uK$. Since it satisfies $N' = 0$, one obtains $C = -Du$ and

$$N' = D(N - uK). \tag{2.14}$$

Frame reciprocity and preservation of the two null directions $N = \pm K$ require

$$K' = D(K - uN). \tag{2.15}$$

Preservation of the interval gives

$$D^2(1 - u^2) = 1. \tag{2.16}$$

Taking the branch continuously connected to the identity,

$$D = \gamma. \tag{2.17}$$

Thus,

$$\boxed{K' = \gamma(K - uN),} \tag{2.18}$$

$$\boxed{N' = \gamma(N - uK).} \tag{2.19}$$

The inverse transformation is

$$K = \gamma(K' + uN'), \qquad N = \gamma(N' + uK'). \tag{2.20}$$

In physical units,

$$\boxed{t' = \gamma\left(t - \frac{vx}{c^2}\right),} \tag{2.21}$$

$$\boxed{x' = \gamma(x - vt).} \tag{2.22}$$

These are the standard Lorentz transformations [1-6].

2.6 Interval invariance

The transformations satisfy

$$-(\Delta K')^2 + (\Delta N')^2 = -(\Delta K)^2 + (\Delta N)^2, \tag{2.23}$$

and therefore

$$\boxed{-c^2\Delta t'^2 + \Delta x'^2 = -c^2\Delta t^2 + \Delta x^2.} \tag{2.24}$$

The proper time along a given trajectory is invariant:

$$\Delta\tau' = \Delta\tau. \tag{2.25}$$

2.7 Velocity transformation and rapidity

For a particle with $w = dN/dK$ in S,

$$w' = \frac{dN\prime}{dK\prime} = \frac{w-u}{1-uw}. \tag{2.26}$$

In physical units,

$$\boxed{V' = \frac{V-v}{1-vV/c^2}.} \tag{2.27}$$

For $V = \pm c$, the transformed velocity remains $\pm c$.

Two successive collinear boosts compose as

$$\boxed{u_{12} = \frac{u_1+u_2}{1+u_1u_2}.} \tag{2.28}$$

Introducing rapidity ξ by $u = \tanh\xi$ gives

$$\gamma = \cosh\xi, \qquad \gamma u = \sinh\xi, \tag{2.29}$$

and successive boosts add linearly,

$$\boxed{\xi_{12} = \xi_1 + \xi_2.} \tag{2.30}$$

2.8 Four-dimensional extension

Define the dimensionless coordinate vector

$$X^A = (K, N^1, N^2, N^3) \tag{2.31}$$

and

$$\eta_{AB} = diag(-1,1,1,1). \tag{2.32}$$

The interval is

$$d\Sigma^2 = \eta_{AB} dX^A dX^B. \tag{2.33}$$

An emergent Lorentz transformation Λ^A_B satisfies

$$\boxed{\Lambda^T \eta \Lambda = \eta.} \tag{2.34}$$

The physical coordinate vector is $x^\mu = (ct, x, y, z)$ with $ct = \ell_P K$ and $x^i = \ell_P N^i$.

2.9 Local Lorentz frames in a gravitational geometry

In a nonuniform grid state,

$$ds^2 = -\alpha^2 c^2 t_P^2 dK^2 + \ell_P^2 h_{ab}(dN^a + \beta^a dK)(dN^b + \beta^b dK). \tag{2.35}$$

At any regular event one may introduce a local orthonormal frame $e^{\hat{A}}_\mu$ satisfying

$$g_{\mu\nu} = \eta_{\hat{A}\hat{B}} e^{\hat{A}}_\mu e^{\hat{B}}_\nu. \tag{2.36}$$

Within a sufficiently small freely falling region, the metric is locally Minkowskian and the laws reduce to the flat-grid Lorentz form. Tidal effects remain through derivatives of the metric.

2.10 Microscopic discreteness and Lorentz symmetry

The continuous transformations generally map integer values of K and N to noninteger values. The full Lorentz group is therefore not generally an exact automorphism of a fixed integer lattice. The primed variables must be interpreted as coarse-grained inertial coordinates.

A generic effective dispersion relation may be written

$$E^2 = p^2c^2 + m^2c^4 + \delta_{grid}(E, p), \tag{2.37}$$

with

$$\lim_{p/p_P \to 0} \delta_{grid} = 0. \tag{2.38}$$

Possible corrections include

$$\delta_{grid} = \eta_1 \frac{p^3c^3}{E_P} + \eta_2 \frac{p^4c^4}{E_P^2} + \cdots. \tag{2.39}$$

Parity, time-reversal invariance, or additional microscopic symmetries may forbid leading terms. The coefficients must be derived and must satisfy observational bounds.

2.11 Illustrative lattice dispersion

A nearest-neighbor discrete wave equation may produce

$$\sin^2\left(\frac{\omega t_P}{2}\right) = \sin^2\left(\frac{k\ell_P}{2}\right) + \frac{\mu^2}{4}. \tag{2.40}$$

For $|\omega t_P| \ll 1$ and $|k\ell_P| \ll 1$,

$$E^2 \simeq p^2c^2 + m^2c^4. \tag{2.41}$$

At higher momenta, the sine functions generate departures. This example illustrates, but does not uniquely determine, the microscopic theory.

2.12 Emergent special-relativistic correspondence theorem

Theorem 1. Suppose that the coarse-grained reference grid satisfies spatial and temporal homogeneity, spatial isotropy, finite isotropic null propagation, linearity of inertial transformations, equivalence of low-energy inertial frames, and invariance of $d\Sigma^2 = -dK^2 + dN^2$. Then the inertial transformations are Lorentz transformations and the emergent observables satisfy time dilation, length contraction, relativity of simultaneity, velocity addition, and

$$E^2 = p^2c^2 + m^2c^4. \tag{2.42}$$

The theorem establishes the relativistic correspondence of the coarse-grained theory. It does not prove that an arbitrary microscopic grid update automatically satisfies its assumptions.

2.13 Relativistic falsification conditions

The relativistic sector would be excluded or severely constrained by any unsuppressed prediction of:

- orientation-dependent local light speed;
- species-dependent limiting velocities;
- failure of measured relativistic time dilation;
- nonuniversal energy-momentum relations;
- vacuum birefringence inconsistent with observation;
- excessive energy-dependent propagation delays;
- composition-dependent free fall;
- directly measurable absolute simultaneity associated with K;
- unstable high-momentum modes;
- large preferred-frame effects in local nongravitational experiments.

The next chapter constructs quantum evolution on the same substrate.

3 Discrete Quantum Evolution and V/M Conversion

3.1 Purpose and scope

The quantum sector is defined fundamentally by a sequence of unitary updates indexed by the universal clock number,

$$\boxed{|\Psi_{K+1}\rangle = U_K|\Psi_K\rangle.} \tag{3.1}$$

The objectives of this chapter are to establish probability-preserving evolution, the continuous-time Schrödinger limit, the relations between phase, action, energy, and momentum, relativistic continuum limits, and a precise effective distinction between extended coherent V states and localized record-bearing M states.

The V and M sectors are not two independently conserved substances. They are effective sectors of a common quantum state distinguished by coherence, localization, dynamical stability, and record formation.

3.2 Hilbert space on the grid

Let the position basis be $\{|i\rangle\}_{i\in\mathcal{V}}$. A single-excitation state is

$$\boxed{|\Psi_K\rangle = \textstyle\sum_{i\in\mathcal{V}} \psi_i\,(K)|i\rangle.} \tag{3.2}$$

Normalization requires

$$\boxed{\langle\Psi_K \mid \Psi_K\rangle = \textstyle\sum_i |\psi_i(K)|^2 = 1.} \tag{3.3}$$

For fields or multiple particles, the Hilbert space is enlarged to a tensor product or Fock-space construction. An effective decomposition is

$$\mathcal{H}_{tot} = \mathcal{H}_V \otimes \mathcal{H}_M \otimes \mathcal{H}_{grid} \otimes \mathcal{H}_{env}. \tag{3.4}$$

This factorization is coarse-grained; all sectors may arise from one microscopic update structure.

3.3 Fundamental unitary update

3.3.1 Postulate 4: Unitary microscopic evolution

For a closed system,

$$|\Psi_{K+1}\rangle = U_K|\Psi_K\rangle, \tag{3.5}$$

with

$$\boxed{U_K^\dagger U_K = U_K U_K^\dagger = I.} \tag{3.6}$$

Therefore,

$$\sum_i |\psi_i(K+1)|^2 = \sum_i |\psi_i(K)|^2. \tag{3.7}$$

After n updates,

$$|\Psi_{K+n}\rangle = U_{K+n-1} \cdots U_K|\Psi_K\rangle. \tag{3.8}$$

For a stationary update rule, $U_K = U$ and $|\Psi_{K+n}\rangle = U^n|\Psi_K\rangle$.

3.4 Locality of the update operator

A local update can be constructed as a product of finite-neighborhood gates,

$$U_K = \prod_\lambda U_{K,\lambda}, \tag{3.9}$$

where each factor acts only on adjacent or finitely separated grid units. A layered gate construction produces an effective causal cone: after n updates, a disturbance can influence only a finite neighborhood. In the reference vacuum,

$$|\Delta N| \leq |\Delta K|. \tag{3.10}$$

Equality corresponds to null propagation.

3.5 Generator and quasi-energy

A unitary operator may be written

$$\boxed{U_K = \exp\left(-\frac{it_P}{\hbar}\hat{H}_K\right),} \tag{3.11}$$

where $\hat{H}_K$ is Hermitian. If

$$U|\varepsilon_n\rangle = e^{-i\theta_n}|\varepsilon_n\rangle, \tag{3.12}$$

the associated quasi-energy is

$$E_n = \frac{\hbar}{t_P}(\theta_n + 2\pi m_n), \qquad m_n \in \mathbb{Z}. \tag{3.13}$$

In the low-energy regime $|E_n t_P/\hbar| \ll 1$, the principal branch is sufficient. If $\hat{H}$ is stationary,

$$U^n = \exp\left(-\frac{int_P}{\hbar}\hat{H}\right). \tag{3.14}$$

With $\tau = nt_P$ in the reference vacuum, this is the ordinary continuous evolution operator.

3.6 Continuous-time limit

For $t_P \parallel \hat{H} \parallel/\hbar \ll 1$,

$$U = I - \frac{it_P}{\hbar}\hat{H} + \mathcal{O}(t_P^2). \tag{3.15}$$

Thus,

$$i\hbar\frac{|\Psi_{K+1}\rangle - |\Psi_K\rangle}{t_P} = \hat{H}|\Psi_K\rangle + \mathcal{O}(t_P). \tag{3.16}$$

In the smooth limit,

$$\boxed{i\hbar\frac{\partial}{\partial\tau}|\Psi(\tau)\rangle = \hat{H}|\Psi(\tau)\rangle.} \tag{3.17}$$

The exact probability conservation belongs to the exponential update; a naive forward-Euler discretization is only an approximation.

3.7 Proper time and local Hamiltonian

In a nontrivial grid state,

$$d\tau = \alpha t_P dK. \tag{3.18}$$

Let $\hat{H}_K$ be the generator per microscopic update and $\hat{H}_\tau$ the generator per unit proper time. Equality of the accumulated phase requires

$$\frac{t_P}{\hbar}\hat{H}_K = \frac{d\tau}{\hbar}\hat{H}_\tau. \tag{3.19}$$

For one update,

$$\boxed{\hat{H}_K = \alpha\hat{H}_\tau,} \qquad \boxed{\hat{H}_\tau = \frac{\hat{H}_K}{\alpha}.} \tag{3.20}$$

A physical observer measures energy with respect to proper time, not directly with respect to the inaccessible microscopic index.

3.8 Phase, action, energy, and momentum

For an energy eigenstate,

$$|E;\tau\rangle = e^{-iE\tau/\hbar}|E;0\rangle. \tag{3.21}$$

The accumulated phase is $-E\tau/\hbar$. More generally,

$$\boxed{\phi = \frac{S}{\hbar}.} \tag{3.22}$$

For two histories,

$$\boxed{\Delta\phi = \frac{\Delta S}{\hbar}.} \tag{3.23}$$

A discrete action may be defined by

$$\boxed{S[\Gamma] = \textstyle\sum_{K=K_i}^{K_f-1} L_K\, t_P,} \tag{3.24}$$

with continuum limit $S = \int L\, d\tau$.

For a stationary phase $e^{-i\omega\tau}$,

$$\boxed{E = \hbar\omega = h\nu.} \tag{3.25}$$

For a spatial translation,

$$\hat{T}(\Delta x) = \exp\left(-\frac{i\Delta x}{\hbar}\hat{p}\right). \tag{3.26}$$

A plane wave

$$\psi_N(K) = A\exp[i(kN\ell_P - \omega K t_P)] \tag{3.27}$$

changes phase by $k\ell_P$ per link, giving

$$\boxed{p = \hbar k = \frac{h}{\lambda}.} \tag{3.28}$$

Energy and momentum are therefore the generators of temporal and spatial phase translations.

The relations

$$E_P t_P = \hbar, \qquad p_P \ell_P = \hbar \tag{3.29}$$

motivate interpreting $\hbar$ as the reference action scale associated with one elementary clock and one elementary grid length. A deterministic large action difference does not by itself cause decoherence; decoherence requires inaccessible fluctuations of relative phase.

3.9 Definition of the V Sector

For a pure state $|\Psi\rangle = \sum_i \psi_i |i\rangle$, define the inverse participation ratio

$$\boxed{IPR = \sum_i |\psi_i|^4,} \tag{3.30}$$

and the effective number of occupied grid units

$$\boxed{N_{eff} = \frac{1}{IPR}.} \tag{3.31}$$

For a density matrix, define an off-diagonal coherence measure

$$\boxed{C_{off} = \sum_{i\neq j} |\rho_{ij}|^2.} \tag{3.32}$$

A V-like state is characterized schematically by

$$N_{eff} \gg 1, \qquad C_{off} \text{ significant}, \qquad L_V \text{ large}. \tag{3.33}$$

Examples include propagating wave packets, coherent radiation, quantum superpositions, and large-scale correlated early-universe configurations.

3.10 Definition of the M Sector

An ideal localized state has $IPR \simeq 1$ and $N_{eff} \simeq 1$. A realistic M state may occupy many microscopic units while remaining localized relative to the corresponding V coherence scale.

The M regime is characterized by

$$N_{eff}^{(M)} \ll N_{eff}^{(V)}, \qquad L_M < L_V, \tag{3.34}$$

and by dynamical persistence,

$$\tau_M \gg \tau_{int}. \tag{3.35}$$

A record is formed when alternatives become correlated with approximately orthogonal environmental states,

$$|\langle E_a \mid E_b\rangle| \ll 1. \tag{3.36}$$

Examples include localized particles, bound states, stable matter configurations, detector records, and macroscopic structures with persistent histories.

3.11 Effective sector decomposition and exchange

Let P_V and P_M be effective projectors satisfying

$$P_V^2 = P_V, \quad P_M^2 = P_M, \quad P_V P_M \simeq 0, \quad P_V + P_M \simeq I. \tag{3.37}$$

The sector weights are

$$\boxed{p_V = Tr(P_V \rho),} \qquad \boxed{p_M = Tr(P_M \rho).} \tag{3.38}$$

The Hamiltonian is decomposed as

$$\hat{H}_{tot} = \hat{H}_V + \hat{H}_M + \hat{H}_{grid} + \hat{H}_{env} + \hat{H}_{int}. \tag{3.39}$$

The interaction mixes the sectors when $P_M \hat{H}_{int} P_V \neq 0$. Under unitary evolution,

$$\frac{d\rho}{d\tau} = -\frac{i}{\hbar}[\hat{H}_{tot}, \rho]. \tag{3.40}$$

The exchange current may be written

$$\boxed{J_{V\to M} = \frac{2}{\hbar} ImTr\left(P_M \hat{H}_{int} P_V \rho\right).} \tag{3.41}$$

Then

$$\frac{dp_M}{d\tau} = J_{V\to M}, \qquad \frac{dp_V}{d\tau} = -J_{V\to M}. \tag{3.42}$$

The current can change sign, so microscopic dynamics permits $V \leftrightarrow M$. Effective irreversibility appears only after correlations disperse into inaccessible environmental degrees of freedom.

If the complete Hamiltonian has no explicit proper-time dependence,

$$\boxed{E_V + E_M + E_{grid} + E_{env} + E_{int} = \text{constant.}} \tag{3.43}$$

Thus, $V \to M$ is not matter creation from nothing; it is redistribution within the complete closed system.

3.12 Entanglement-induced decoherence

For an initial superposition

$$(c_1|1\rangle + c_2|2\rangle)|E_0\rangle, \tag{3.44}$$

unitary interaction produces

$$|\Psi\rangle = c_1|1\rangle|E_1\rangle + c_2|2\rangle|E_2\rangle. \tag{3.45}$$

The reduced density matrix contains off-diagonal terms multiplied by

$$\boxed{\Gamma = \langle E_2 \mid E_1\rangle.} \tag{3.46}$$

When $|\Gamma| \ll 1$, local interference is suppressed while the total state remains pure and unitary [12,13]. If the relative action fluctuates approximately Gaussianly,

$$\Gamma = \left\langle e^{i\Delta S/\hbar}\right\rangle = \exp\left[\frac{i\langle\Delta S\rangle}{\hbar} - \frac{Var(\Delta S)}{2\hbar^2}\right]. \tag{3.47}$$

Hence,

$$\boxed{|\Gamma| = e^{-\mathcal{D}}, \qquad \mathcal{D} = \frac{Var(\Delta S)}{2\hbar^2}.} \tag{3.48}$$

A representative reduced master equation is

$$\boxed{\frac{d\rho}{d\tau} = -\frac{i}{\hbar}[\hat{H},\rho] - \Lambda\left[\hat{X},[\hat{X},\rho]\right].} \tag{3.49}$$

In the position basis, spatially separated coherences decay as

$$\rho(x,x';\tau) \propto \exp[-\Lambda(x-x')^2\tau]. \tag{3.50}$$

3.13 Effective V-to-M Conversion Criterion

Decoherence alone is insufficient to define a stable M state. The minimum conditions are

$$\boxed{\mathcal{D} \gtrsim 1, \qquad L_M < L_V, \qquad M \text{ locally stable,} \qquad \tau_{record} \gg \tau_{int}.} \tag{3.51}$$

A local collective coordinate q must satisfy

$$\left.\frac{\partial E_{eff}}{\partial q}\right|_{q_M} = 0, \qquad \left.\frac{\partial^2 E_{eff}}{\partial q^2}\right|_{q_M} > 0. \tag{3.52}$$

Recoherence must also be dynamically inaccessible on the observational timescale.

3.14 Measurement, collapse, and the Born rule

Measurement is represented as record formation,

$$\left(\textstyle\sum_i c_i\,|s_i\rangle\right)|A_0\rangle|E_0\rangle \to \textstyle\sum_i c_i\,|s_i\rangle|A_i\rangle|E_i\rangle, \tag{3.53}$$

with approximately orthogonal apparatus and environmental records. Decoherence explains suppression of interference, pointer-basis selection, and persistent classical alternatives. It does not by itself select one unique term from the total superposition.

The standard probability rule

$$\boxed{P(i) = |c_i|^2} \tag{3.54}$$

is retained as a quantum correspondence postulate. Unitarity guarantees conservation of $\sum_i |c_i|^2$, but not a complete derivation of the probability interpretation or individual outcome selection.

3.15 Information conservation and irreversibility

For a closed unitary system, the total von Neumann entropy

$$S_{vN} = -k_B Tr(\rho \ln \rho) \tag{3.55}$$

is constant. A subsystem entropy may increase because of entanglement with degrees of freedom that are traced out. Thermodynamic entropy is therefore interpreted as information inaccessible under a specified coarse-graining. Microscopic reversibility is compatible with macroscopic practical irreversibility because recurrence or recoherence requires control over an enormous number of dispersed correlations.

3.16 Effective field representation

At coarse-grained scales, representative order parameters $V(x)$ and $M(x)$ may be introduced through

$$S_{VM} = \int d^4x\sqrt{-g}(\mathcal{L}_V + \mathcal{L}_M + \mathcal{L}_{int}). \tag{3.56}$$

A minimum interaction is

$$\boxed{\mathcal{L}_{int} = -\lambda(\chi)|V|^2|M|^2.} \tag{3.57}$$

A direct conversion term may be written

$$\boxed{\mathcal{L}_{conv} = -g(\chi)\left(M^\dagger V + V^\dagger M\right),} \tag{3.58}$$

when compatible with the relevant symmetries and quantum numbers. These terms are effective and will be embedded in the cosmological action.

3.17 Quantum correspondence theorems

Theorem 2 (Schrödinger continuum correspondence). If the microscopic evolution is local and unitary, admits a Hermitian low-energy generator, contains a graph kinetic operator, and varies slowly over one link and one update, then the coarse-grained wavefunction satisfies the Schrödinger equation.

Theorem 3 (Unitary V/M probability exchange). If P_V and P_M are time-independent orthogonal projectors with $P_V + P_M = I$ and the total density operator evolves unitarily, then

$$\boxed{\frac{dp_V}{d\tau} + \frac{dp_M}{d\tau} = 0.} \tag{3.59}$$

3.18 Quantum-sector falsification conditions

The quantum sector would be excluded or severely constrained by probability nonconservation in closed systems, incorrect $E = \hbar\omega$ or $p = \hbar k$, ordinary-energy departures from quantum interference, species-dependent $\hbar$, uncontrolled superluminal propagation, spontaneous energy creation in V/M conversion, or claims of outcome selection unsupported by the microscopic dynamics.

The next chapter introduces the grid-state variables and develops gravity as an emergent response of the lapse, shift, and spatial metric.

4 Processing Load and Emergent Gravity

4.1 Conceptual status

In Grid-Clock Cosmology, gravity is not primarily an additional force transmitted between particles inside a pre-existing spacetime. It is interpreted as a change in the effective lapse, shift, and spatial metric generated by the dynamical state of the underlying grid. The hierarchy is

$$\boxed{\text{grid-clock substrate} \supset \text{emergent spacetime} \supset \text{quantum fields} \supset \text{particles and matter.}} \tag{4.1}$$

Gravity belongs to the level at which physical time, distance, and causal propagation are defined. Electromagnetic and nuclear interactions operate primarily between fields and particles after spacetime has emerged.

The central map is

$$\boxed{\text{local grid state} \to (\alpha, \beta^a, h_{ab}) \to g_{\mu\nu}.} \tag{4.2}$$

The full dynamics are constructed from an effective action in the next chapter. Here we establish the minimum field content, the weak-field correspondence, and the observational requirements.

4.2 Gravity requires scalar, vector, and tensor grid information

A single scalar processing-load variable can describe gravitational clock-rate variations and the scalar part of weak fields, but cannot reproduce a general metric. The microscopic grid state must contain at least three coarse-grained classes of information:

$$\boxed{\chi, \qquad J^a, \qquad \Pi^{ab}.} \tag{4.3}$$

Here χ is a scalar grid-state variable, J^a is a directional update current, and Π^{ab} is an anisotropic grid-state tensor. Their effective geometric images are

$$\chi \to \alpha \text{ and scalar } h_{ab}, \tag{4.4}$$

$$J^a \to \beta^a, \tag{4.5}$$

$$\Pi^{ab} \to \text{anisotropic and tensor parts of } h_{ab}. \tag{4.6}$$

The scalar χ is the minimum variable for the homogeneous cosmological and static weak-field sectors; it is not the complete gravitational field.

4.3 Operational definition of processing load

The term *processing load* does not mean the workload of a conventional digital computer. It denotes the local dynamical activity required to maintain and update a microscopic grid configuration relative to the reference vacuum.

Let $\hat{C}_i$ be a local Hermitian operator measuring an appropriate update activity. A dimensionless coarse-grained scalar may be defined by

$$\boxed{\chi_i(K) = \frac{\langle \hat{C}_i \rangle_K - C_0}{C_*},} \tag{4.7}$$

where C_0 is the reference-vacuum value and C_* is a characteristic response scale. In the continuum approximation, $\chi_i(K) \to \chi(x)$ and the reference vacuum is $\chi = 0$.

The directional current and anisotropic stress are defined schematically by normalized expectation values of local update-current and update-stress operators. Their exact microscopic forms remain open.

4.4 Constitutive map to geometry

The emergent variables are functionals of the coarse-grained state:

$$\boxed{\alpha = \mathcal{A}[\chi, J, \Pi],} \tag{4.8}$$

$$\boxed{\beta^a = \mathcal{B}^a[\chi, J, \Pi],} \tag{4.9}$$

$$\boxed{h_{ab} = \mathcal{H}_{ab}[\chi, J, \Pi].} \tag{4.10}$$

For a static isotropic scalar configuration, $J^a = 0$ and $\Pi^{ab} = 0$, so

$$\alpha = \alpha(\chi), \qquad h_{ab} = b^2(\chi)\delta_{ab}. \tag{4.11}$$

The reference-vacuum normalization is

$$\boxed{\alpha(0) = 1, \qquad b(0) = 1.} \tag{4.12}$$

A regular causal and spatial geometry requires $\alpha^2 > 0$ and $b^2 > 0$ on the physical branch.

4.5 ADM-like physical metric

With reference physical coordinates $t = t_P K$ and $x^i = \ell_P N^i$, the line element is

$$\boxed{ds^2 = -\alpha^2 c^2 dt^2 + h_{ij}\left(dx^i + c\beta^i dt\right)\left(dx^j + c\beta^j dt\right).} \tag{4.13}$$

The lapse controls proper time, the shift describes directional flow of the spatial grid relative to constant-K hypersurfaces, and the spatial metric controls physical lengths and angles. For a static observer with negligible shift,

$$\boxed{d\tau = \alpha\, dt.} \tag{4.14}$$

For neighboring points at equal t,

$$\boxed{d\ell^2 = h_{ij} dx^i dx^j.} \tag{4.15}$$

4.6 Weak-field scalar expansion

For $|\chi| \ll 1$, adopt the minimum expansion

$$\boxed{\alpha^2(\chi) = 1 - \chi + \mathcal{O}(\chi^2),} \tag{4.16}$$

and

$$\boxed{h_{ij} = \left[1 + \gamma_g \chi + \mathcal{O}(\chi^2)\right]\delta_{ij}.} \tag{4.17}$$

Define the Newtonian potential by

$$\boxed{\chi = -\frac{2\Phi}{c^2}.} \tag{4.18}$$

For positive isolated mass, $\Phi < 0$ and therefore $\chi > 0$. The metric becomes

$$\boxed{ds^2 = -\left(1 + \frac{2\Phi}{c^2}\right)c^2 dt^2 + \left(1 - \frac{2\gamma_g \Phi}{c^2}\right)d\mathbf{x}^2.} \tag{4.19}$$

General relativity without significant anisotropic stress requires

$$\boxed{\gamma_g = 1.} \tag{4.20}$$

The earlier heuristic relations $\alpha^2 \simeq 1 - \chi$ and $b^2 \simeq 1 + \chi$ are recovered at this value. They are weak-field correspondence relations and must not be extrapolated unchanged into strong fields.

4.7 Gravitational time dilation and redshift

For a static observer,

$$d\tau = \alpha dt \simeq \left(1 + \frac{\Phi}{c^2}\right)dt = \left(1 - \frac{\chi}{2}\right)dt. \tag{4.21}$$

A clock deeper in an attractive potential accumulates less proper time per reference time. For two clocks,

$$\boxed{\frac{d\tau_1}{d\tau_2} \simeq 1 + \frac{\Phi_1 - \Phi_2}{c^2}.} \tag{4.22}$$

The locally measured photon frequency is inversely proportional to the lapse. For emission at e and observation at o,

$$\boxed{\frac{\omega_o}{\omega_e} = \frac{\alpha_e}{\alpha_o},} \tag{4.23}$$

and

$$\boxed{1 + z_{grav} = \frac{\alpha_o}{\alpha_e}.} \tag{4.24}$$

In the weak field,

$$z_{grav} \simeq \frac{\Phi_o - \Phi_e}{c^2} \simeq \frac{\chi_e - \chi_o}{2}. \tag{4.25}$$

This redshift is geometric; it need not be interpreted as frictional energy loss by the photon.

4.8 Newtonian particle motion

Matter is assumed to couple universally to the emergent metric. The point-particle action is

$$\boxed{S_p = -mc^2 \int d\tau.} \tag{4.26}$$

For $v \ll c$ and $|\Phi| \ll c^2$,

$$\frac{d\tau}{dt} \simeq 1 + \frac{\Phi}{c^2} - \frac{v^2}{2c^2}. \tag{4.27}$$

The Lagrangian is

$$L_p \simeq -mc^2 - m\Phi + \frac{1}{2}mv^2. \tag{4.28}$$

Dropping the constant rest-energy term gives

$$\boxed{L_{NR} = \frac{1}{2}mv^2 - m\Phi.} \tag{4.29}$$

The Euler-Lagrange equation yields

$$\boxed{\frac{d^2\mathbf{x}}{dt^2} = -\nabla\Phi.} \tag{4.30}$$

Using $\Phi = -c^2\chi/2$,

$$\boxed{\frac{d^2\mathbf{x}}{dt^2} = \frac{c^2}{2}\nabla\chi.} \tag{4.31}$$

For an ordinary source, χ increases toward the mass and the acceleration is attractive.

4.9 Geodesic interpretation of inertia and gravity

In a homogeneous reference grid, a stable update pattern satisfies $\Delta_K u = 0$. In a nonuniform grid state, the propagation pattern follows an extremum of proper time,

$$\delta \int d\tau = 0. \tag{4.32}$$

The resulting equation is

$$\boxed{\frac{d^2x^\mu}{d\tau^2} + \Gamma^\mu_{\alpha\beta}\frac{dx^\alpha}{d\tau}\frac{dx^\beta}{d\tau} = 0.} \tag{4.33}$$

Thus, free fall is not an ordinary force acting on an unchanged background. It is inertial propagation in a nonuniform emergent geometry.

4.10 Weak-field source equation

Newtonian correspondence requires

$$\boxed{\nabla^2\Phi = 4\pi G\rho.} \tag{4.34}$$

Using $\chi = -2\Phi/c^2$ gives

$$\boxed{\nabla^2\chi = -\frac{8\pi G}{c^2}\rho.} \tag{4.35}$$

This fixes the normalization and sign of the scalar response in the weak-field regime. At the relativistic level the source must be the complete stress-energy tensor, not rest mass alone.

For a point mass,

$$\Phi(r) = -\frac{GM}{r}, \tag{4.36}$$

and therefore

$$\boxed{\chi(r) = \frac{2GM}{c^2 r}.} \tag{4.37}$$

The lapse and spatial metric reproduce the first-order Schwarzschild field in isotropic coordinates when $\gamma_g = 1$. The acceleration is

$$\boxed{\mathbf{a} = -\frac{GM}{r^2}\hat{\mathbf{r}}.} \tag{4.38}$$

4.11 Universal coupling and the equivalence principle

Let the matter action depend on χ only through the common metric at leading order,

$$\boxed{S_{matter} = S_{matter}[\psi_A, g_{\mu\nu}(\chi, J, \Pi)].} \tag{4.39}$$

Species-dependent direct couplings would generally produce composition-dependent free fall. For a point particle, the mass factor cancels from the geodesic equation, so all test bodies with the same initial conditions follow the same trajectory. In the Newtonian limit,

$$\boxed{m_{inertial} = m_{gravitational}.} \tag{4.40}$$

The equivalence is not interpreted as a coincidence between two charges; it follows from universal propagation in one geometry.

At any regular event a freely falling local frame can be chosen with

$$g_{\hat{\mu}\hat{\nu}} = \eta_{\hat{\mu}\hat{\nu}}, \qquad \partial_{\hat{\alpha}} g_{\hat{\mu}\hat{\nu}} = 0. \tag{4.41}$$

Gravity cannot be removed over an extended region when the Riemann tensor is nonzero. Measurable gravity is associated with spatial variation and tidal curvature.

4.12 Light deflection and Shapiro delay

The lapse response alone reproduces slow-particle acceleration, but correct light deflection also requires the spatial metric. Write

$$ds^2 = -\left(1 + \frac{2\Phi}{c^2}\right)c^2 dt^2 + \left(1 - \frac{2\Psi}{c^2}\right)d\mathbf{x}^2, \tag{4.42}$$

with

$$\Psi = \gamma_g \Phi. \tag{4.43}$$

The leading lensing deflection is

$$\boxed{\hat{\boldsymbol{\alpha}}_{lens} = -\frac{1}{c^2}\int \nabla_\perp(\Phi + \Psi)\, d\ell.} \tag{4.44}$$

For $\Psi = \Phi$ and a point mass with impact parameter b,

$$\boxed{|\hat{\alpha}_{lens}| = \frac{4GM}{bc^2}.} \tag{4.45}$$

The Shapiro delay is

$$\boxed{\Delta t_{Shapiro} = -\frac{1}{c^3}\int (\Phi + \Psi)\, d\ell.} \tag{4.46}$$

These tests strongly constrain the relation between clock-rate and spatial responses.

4.13 Shift, gravitomagnetism, and rotation

Moving or rotating sources require a directional update current J^i mapped to the shift vector,

$$\boxed{\beta^i = \mathcal{B}^i[J, \chi, \Pi].} \tag{4.47}$$

In the weak field, the metric contains a gravitomagnetic potential $\mathbf{A}_g$,

$$ds^2 \simeq -\left(1 + \frac{2\Phi}{c^2}\right)c^2 dt^2 - \frac{4}{c}\mathbf{A}_g \cdot d\mathbf{x}\, dt + \left(1 - \frac{2\gamma_g \Phi}{c^2}\right)d\mathbf{x}^2. \tag{4.48}$$

A rotating source produces nonzero grid flow and observable frame dragging, gyroscope precession, and orbital-node precession.

4.14 Anisotropic response and gravitational waves

Decompose the spatial metric as

$$\boxed{h_{ij} = a^2 e^{2\zeta_g} (e^{2q})_{ij}, \qquad q^i{}_i = 0.} \tag{4.49}$$

For small perturbations,

$$h_{ij} \simeq a^2\left[\left(1 + 2\zeta_g\right)\delta_{ij} + 2q_{ij}\right]. \tag{4.50}$$

The transverse-traceless part satisfies

$$\partial^i q_{ij}^{TT} = 0, \qquad \delta^{ij} q_{ij}^{TT} = 0. \tag{4.51}$$

In the local low-energy limit, a viable theory must reproduce

$$\boxed{\partial_t^2 q_{ij}^{TT} - c_T^2 \nabla^2 q_{ij}^{TT} = \mathcal{S}_{ij}^{TT},} \tag{4.52}$$

with

$$\boxed{c_T = c} \tag{4.53}$$

to observational accuracy [49]. In an expanding universe,

$$\ddot{q}_{ij}^{TT} + 3H\dot{q}_{ij}^{TT} - \frac{c_T^2}{a^2}\nabla^2 q_{ij}^{TT} = \mathcal{S}_{ij}^{TT}. \tag{4.54}$$

A scalar processing-load variable must not eliminate the tensor degrees of freedom required for gravitational radiation.

4.15 Why ordinary matter does not shield gravity

Electromagnetic fields can be screened by rearranging positive and negative charges. The emergent metric is not sourced by an analogous positive and negative gravitational charge. It is defined throughout the grid, including the units supporting matter. Matter does not need to be penetrated by a stream of gravitational particles; the material body itself is part of the geometry-generating state. No general shielding follows from universal metric coupling.

4.16 Conservation of energy and momentum

The complete action includes gravity, the grid field, the V and M sectors, matter, and interactions. The total stress-energy tensor satisfies

$$\boxed{\nabla_\mu T^{\mu\nu}_{total} = 0.} \tag{4.55}$$

Individual sectors may exchange energy and momentum,

$$\nabla_\mu T_A^{\mu\nu} = Q_A^\nu, \tag{4.56}$$

with

$$\boxed{\textstyle\sum_A Q_A^\nu = 0.} \tag{4.57}$$

An apparent loss from matter or radiation must be balanced by the grid or interaction sectors. A local, gauge-independent gravitational energy density is not generally defined in arbitrary coordinates; the meaningful statement is conservation of the complete covariant system and appropriate global charges when symmetries permit them.

4.17 Cosmological background, local gravity, and direction dependence

Decompose

$$\boxed{\chi(x) = \bar{\chi}(t) + \delta\chi(t, \mathbf{x}).} \tag{4.58}$$

The homogeneous component contributes to background expansion, while the perturbation contributes to local gravitational potentials and structure formation. The scale factor $a(t)$ must not be counted again as a local response $b(\chi)$.

A homogeneous anisotropic solution may be written

$$h_{ij} = a_i^2(t)\delta_{ij}, \qquad a_i = ae^{s_i}, \qquad \sum_i s_i = 0. \tag{4.59}$$

The directional Hubble rates are

$$H_i = H + \dot{s}_i, \tag{4.60}$$

and the shear is $\sigma^i{}_j = \dot{s}_i\delta^i{}_j$. Along a line of sight,

$$\boxed{H_\parallel = H + \sigma_{ij}n^i n^j.} \tag{4.61}$$

A vector grid mode may add a dipole. A minimum phenomenological expression is

$$\boxed{H(z,\hat{\mathbf{n}}) \simeq H_{iso}(z)\left[1 + A_1(z)\hat{\mathbf{n}}\cdot\hat{\mathbf{p}} + A_2(z)Q_{ij}n^i n^j\right].} \tag{4.62}$$

The framework permits directional modes; it does not yet require nonzero amplitudes.

4.18 Bounded response and strong fields

If the grid has a finite local update capacity, a normalized load fraction may approach a limiting value. The geometry may respond nonlinearly as the limit is approached. However, the weak-field expression $\alpha^2 \simeq 1 - \chi$ does not establish the exact strong-field relation $\alpha^2 = 1 - \chi$. Identifying $\chi = 1$ automatically with a horizon would be unjustified.

A generic nonlinear scalar equation may take the form

$$\boxed{\nabla_i\left[\mu_\chi(\chi, X)\nabla^i\chi\right] - \frac{dU_\chi}{d\chi} = \mathcal{S}_\chi,} \tag{4.63}$$

where $X = \frac{1}{2}\nabla_i\chi\nabla^i\chi$. The Newtonian limit requires $\mu_\chi \to 1$, $dU_\chi/d\chi \to 0$, and $\mathcal{S}_\chi \to -8\pi G\rho/c^2$. Strong-field behavior must be derived from the nonlinear action.

4.19 Weak-field theorems

Theorem 4 (Newtonian correspondence). Universal metric coupling, $\alpha^2 = 1 - \chi + \mathcal{O}(\chi^2)$, $\chi = -2\Phi/c^2$, and $\nabla^2\chi = -8\pi G\rho/c^2$ imply

$$\frac{d^2\mathbf{x}}{dt^2} = -\nabla\Phi, \qquad \nabla^2\Phi = 4\pi G\rho. \tag{4.64}$$

Theorem 5 (Universal free fall). If all test bodies have action $S_A = -m_A c\int ds$ without leading species-dependent direct coupling to χ, the factor m_A cancels from the equations of motion and all bodies with the same initial conditions follow the same trajectory.

Theorem 6 (Local invariance). If all nongravitational fields couple universally to a smooth metric and microscopic corrections are negligible, a freely falling frame exists in which the leading local equations reduce to their special-relativistic form.

4.20 Gravitational falsification conditions

The emergent-gravity interpretation would be excluded or severely constrained by incorrect Newtonian acceleration, tested inverse-square-law violations, composition-dependent free fall, nonuniversal redshift, an incorrect lensing-to-time-dilation relation, gravitational-wave speed different from c, large unobserved polarizations, conventional gravitational shielding, ghosts or unstable modes, excessive preferred-frame effects, or an undefined strong-field metric.

The next chapter constructs the covariant effective action.

5 Effective Action and the Cosmological Correspondence Sector

5.1 Purpose of the action formulation

The preceding chapters defined the kinematical and low-energy correspondence structure,

$$(\mathcal{G},K)\rightarrow\left(\alpha,\beta^i,h_{ij}\right)\rightarrow g_{\mu\nu}, \tag{5.1}$$

and the discrete unitary quantum law

$$|\Psi_{K+1}\rangle=U_K|\Psi_K\rangle. \tag{5.2}$$

A physical theory requires a dynamical principle from which field equations, conservation laws, cosmological evolution, and perturbations follow. The minimum effective action must recover Einstein gravity in its tested regime, couple all matter universally to one metric, contain a dynamical grid-state field, describe the effective V and M sectors and their exchange, admit an exact ΛCDM correspondence point, and possess stable perturbations.

The action developed here is a low-energy effective action. It is not yet derived uniquely from the microscopic graph update operator.

5.2 Conventions

The metric signature is $(-,+,+,+)$ and the reduced Planck mass is

$$\boxed{M_{Pl}^2=\frac{1}{8\pi G}}. \tag{5.3}$$

The Einstein-Hilbert action is

$$\boxed{S_{EH}=\int d^4x\sqrt{-g}\,\frac{M_{Pl}^2}{2}R.} \tag{5.4}$$

For a dimensionless scalar grid field χ, define

$$\boxed{X_\chi=-\frac{1}{2}g^{\mu\nu}\nabla_\mu\chi\nabla_\nu\chi.} \tag{5.5}$$

For a homogeneous field in a Friedmann universe, $X_\chi=\dot{\chi}^2/2$.

5.3 Minimum covariant action

The baseline action is

$$\boxed{S=S_{grav}+S_G+S_V+S_M+S_{int}+S_b+S_r+S_{dir}.} \tag{5.6}$$

Explicitly,

$$\boxed{S=\int d^4x\sqrt{-g}\left[\frac{M_{Pl}^2}{2}R+P_G\left(\chi,X_\chi\right)+\mathcal{L}_V+\mathcal{L}_M\qquad+\mathcal{L}_{int}+\mathcal{L}_{dir}\right]+S_b\left[g_{\mu\nu},\psi_b\right]+S_r\left[g_{\mu\nu},\psi_r\right].} \tag{5.7}$$

Here P_G is the effective grid-field Lagrangian, $\mathcal{L}_V$ describes coherent extended degrees of freedom, $\mathcal{L}_M$ describes localized record-bearing degrees of freedom, $\mathcal{L}_{int}$ generates exchange, and $\mathcal{L}_{dir}$ contains optional directional modes.

The baseline theory is minimally coupled in the Einstein frame. Ordinary matter sees the same metric $g_{\mu\nu}$, which is essential for the equivalence principle.

5.4 Why the baseline gravitational term is Einstein-Hilbert

A general scalar-tensor action could contain $F(\chi)R$. Such a term generally produces a variable gravitational coupling and additional fifth-force effects. The minimum correspondence model adopts

$$\boxed{F(\chi)=1} \tag{5.8}$$

over the tested regime. New physics is introduced through the grid-sector stress-energy and interactions rather than by arbitrarily changing the tensor structure of gravity at the outset.

5.5 General grid-field Lagrangian

The grid field is described by

$$\boxed{\mathcal{L}_G = P_G(\chi, X_\chi).} \tag{5.9}$$

This includes canonical scalar fields, k-essence, nonlinear kinetic fields, and effective fluid descriptions [35-40]. The canonical special case is

$$\boxed{P_G = Z_G(\chi)X_\chi - U_G(\chi).} \tag{5.10}$$

A canonical field φ may be introduced by $d\varphi/d\chi = \sqrt{Z_G(\chi)}$. The general P_G form is retained because a canonical scalar has relativistic sound speed and cannot automatically behave as clustering cold dark matter.

5.6 Stress-energy and equation of motion

Variation with respect to the metric gives

$$\boxed{T_{\mu\nu}^{(G)} = P_{G,X}\nabla_\mu\chi\nabla_\nu\chi + P_G g_{\mu\nu}.} \tag{5.11}$$

For timelike $\nabla_\mu\chi$, define

$$u_\mu^{(G)} = \frac{\nabla_\mu\chi}{\sqrt{2X_\chi}}. \tag{5.12}$$

Then

$$T_{\mu\nu}^{(G)} = (\rho_G + p_G)u_\mu u_\nu + p_G g_{\mu\nu}, \tag{5.13}$$

with

$$\boxed{p_G = P_G,} \tag{5.14}$$

$$\boxed{\rho_G = 2X_\chi P_{G,X} - P_G.} \tag{5.15}$$

The equation-of-state parameter is $w_G = p_G/\rho_G$.

For the canonical form,

$$\rho_G = Z_G X_\chi + U_G, \qquad p_G = Z_G X_\chi - U_G, \tag{5.16}$$

so

$$\boxed{w_G = \frac{Z_G X_\chi - U_G}{Z_G X_\chi + U_G}.} \tag{5.17}$$

Matter-like behavior requires approximately $Z_G X_\chi \simeq U_G$, while vacuum-like behavior requires $Z_G X_\chi \ll U_G$.

Variation with respect to χ gives

$$\boxed{\nabla_\mu(P_{G,X}\nabla^\mu\chi) + P_{G,\chi} = \mathcal{S}_\chi^{int}.} \tag{5.18}$$

For a homogeneous field,

$$\boxed{(P_{G,X} + 2X_\chi P_{G,XX})\ddot{\chi} + 3HP_{G,X}\dot{\chi} + 2X_\chi P_{G,X\chi} - P_{G,\chi} = \mathcal{S}_\chi.} \tag{5.19}$$

For $P_G = Z_G X_\chi - U_G$,

$$Z_G(\ddot{\chi} + 3H\dot{\chi}) + \frac{1}{2}Z_{G,\chi}\dot{\chi}^2 + U_{G,\chi} = \mathcal{S}_\chi. \tag{5.20}$$

5.7 Metric field equation

The complete variation gives

$$\boxed{M_{Pl}^2 G_{\mu\nu} = T_{\mu\nu}^{(b)} + T_{\mu\nu}^{(r)} + T_{\mu\nu}^{(G)} + T_{\mu\nu}^{(V)} + T_{\mu\nu}^{(M)} + T_{\mu\nu}^{(int)} + T_{\mu\nu}^{(dir)}.} \tag{5.21}$$

Thus, the geometrical tensor structure is that of general relativity in the baseline model. The new physical content lies in the grid-sector stress tensor and its interactions.

5.8 Effective V and M Order Parameters

The symbols V and M denote dynamical sectors rather than elementary scalar particles. For an effective field description introduce representative order parameters $\mathcal{V}(x)$ and $\mathcal{M}(x)$,

$$\boxed{\mathcal{L}_V = -\frac{1}{2}Z_V(\chi)g^{\mu\nu}\nabla_\mu\mathcal{V}\nabla_\nu\mathcal{V} - W_V(\mathcal{V},\chi),} \tag{5.22}$$

$$\boxed{\mathcal{L}_M = -\frac{1}{2}Z_M(\chi)g^{\mu\nu}\nabla_\mu\mathcal{M}\nabla_\nu\mathcal{M} - W_M(\mathcal{M},\chi).} \tag{5.23}$$

A simple interaction is

$$\boxed{\mathcal{L}_{int} = -\lambda_{VM}(\chi)\mathcal{V}^2\mathcal{M}^2.} \tag{5.24}$$

A direct conversion term may be written

$$\boxed{\mathcal{L}_{conv} = -g_{VM}(\chi)\mathcal{V}\mathcal{M},} \tag{5.25}$$

when permitted by symmetries and conserved quantum numbers. The field $\mathcal{M}$ is an order parameter for the localized cosmological sector; it does not replace the detailed particle content of ordinary matter.

5.9 Energy exchange between sectors

Individual sector stress tensors need not be conserved separately,

$$\boxed{\nabla_\mu T^{\mu\nu}_{(A)} = Q^\nu_A,} \tag{5.26}$$

while total covariance requires

$$\boxed{\textstyle\sum_A Q^\nu_A = 0.} \tag{5.27}$$

For a homogeneous cosmology,

$$\boxed{\dot{\rho}_A + 3H(\rho_A + p_A) = Q_A.} \tag{5.28}$$

The total energy density satisfies the ordinary conservation equation. A conversion $V \to M$ transfers energy and correlations between sectors; it does not violate conservation.

5.10 Homogeneous cosmological reduction

For a flat Friedmann geometry,

$$ds^2 = -c^2dt^2 + a^2(t)d\mathbf{x}^2. \tag{5.29}$$

The Friedmann equations are

$$\boxed{3M_{Pl}^2H^2 = \rho_r + \rho_b + \rho_G + \rho_V + \rho_M + \rho_{dir},} \tag{5.30}$$

and

$$\boxed{-2M_{Pl}^2\dot{H} = \textstyle\sum_A(\rho_A + p_A).} \tag{5.31}$$

Equivalently,

$$\frac{\ddot{a}}{a} = -\frac{1}{6M_{Pl}^2}\sum_A(\rho_A + 3p_A). \tag{5.32}$$

Accelerated expansion occurs when $\rho_{total} + 3p_{total} < 0$.

For the minimum observational model, the grid-related sectors are grouped into one effective density ρ_G and pressure p_G.

5.11 Target equation of state and density

The phenomenological target is

$$\boxed{w_G(a) = -\frac{1}{1+(a_t/a)^n}.} \tag{5.33}$$

At $a \ll a_t$, $w_G \simeq 0$; at $a \gg a_t$, $w_G \simeq -1$.

Conservation,

$$\dot{\rho}_G + 3H(1+w_G)\rho_G = 0, \tag{5.34}$$

gives

$$\boxed{\rho_G(a) = \rho_{G0} a^{-3}\left[\frac{a^n+a_t^n}{1+a_t^n}\right]^{3/n}.} \tag{5.35}$$

Equivalently,

$$\boxed{\rho_G(a) = \rho_{G0}\left[\frac{1+a_t^n a^{-n}}{1+a_t^n}\right]^{3/n}.} \tag{5.36}$$

5.12 Barotropic Representation and the Special Value n = 3

Define

$$A_s = \frac{1}{1+a_t^n}, \qquad \alpha_B = \frac{n}{3} - 1. \tag{5.37}$$

Then

$$\rho_G(a) = \rho_{G0}\left[A_s + (1-A_s)a^{-3(1+\alpha_B)}\right]^{1/(1+\alpha_B)}, \tag{5.38}$$

which corresponds at the homogeneous level to

$$\boxed{p_G = -\frac{A}{\rho_G^{\alpha_B}}.} \tag{5.39}$$

For

$$\boxed{n = 3,} \tag{5.40}$$

we have $\alpha_B = 0$, constant pressure, and

$$\rho_G(a) = \rho_{G0}\frac{1+a_t^3 a^{-3}}{1+a_t^3}. \tag{5.41}$$

Define

$$\boxed{\rho_\Lambda = \frac{\rho_{G0}}{1+a_t^3},} \tag{5.42}$$

$$\boxed{\rho_{c0} = \frac{\rho_{G0} a_t^3}{1+a_t^3}.} \tag{5.43}$$

Then

$$\boxed{\rho_G(a) = \rho_\Lambda + \rho_{c0} a^{-3}.} \tag{5.44}$$

Thus, $n = 3$ is the exact ΛCDM background limit, not merely an approximation. In density parameters,

$$\Omega_c = \Omega_G\frac{a_t^3}{1+a_t^3}, \qquad \Omega_\Lambda = \Omega_G\frac{1}{1+a_t^3}. \tag{5.45}$$

5.13 V/M Interpretation of n

Decompose the effective grid density into pressureless matter-like and vacuum-like sectors,

$$\rho_G = \rho_M + \rho_V, \qquad p_M = 0, \qquad p_V = -\rho_V. \tag{5.46}$$

Then

$$w_G = -\frac{\rho_V}{\rho_M+\rho_V}, \tag{5.47}$$

so

$$\boxed{\frac{\rho_M}{\rho_V} = \left(\frac{a_t}{a}\right)^n.} \tag{5.48}$$

Let

$$\dot{\rho}_M + 3H\rho_M = Q, \qquad \dot{\rho}_V = -Q. \tag{5.49}$$

The required exchange rate is

$$\boxed{Q = (3-n)H\frac{\rho_M\rho_V}{\rho_M+\rho_V}.} \tag{5.50}$$

Therefore:

- $n = 3$ implies $Q = 0$ and exact noninteracting cold matter plus vacuum;
- $n < 3$ gives $Q > 0$, transfer from the vacuum-like sector toward the matter-like sector in the adopted convention;
- $n > 3$ gives $Q < 0$, transfer from the matter-like sector toward the vacuum-like sector.

The interaction Lagrangian must ultimately generate this law or a more fundamental relation from which it follows. The form must be fixed before validation data are examined.

5.14 Linear perturbation requirements

In Newtonian gauge,

$$\boxed{ds^2 = -(1+2\Phi)c^2dt^2 + a^2(t)(1-2\Psi)d\mathbf{x}^2.} \tag{5.51}$$

A complete perturbation model requires the background w_G, physical sound speed, anisotropic stress, perturbed exchange δQ, and momentum-transfer prescription. A viable early matter-like regime requires

$$w_G \simeq 0, \qquad c_{s,G}^2 \simeq 0, \qquad \pi_G \simeq 0. \tag{5.52}$$

The minimum clustering fraction is

$$\boxed{\rho_{G,cl} \simeq (1+w_G)\rho_G = \rho_M.} \tag{5.53}$$

Thus,

$$\boxed{\Omega_{cl}(a) = \frac{\rho_b+(1+w_G)\rho_G}{3M_{Pl}^2H^2}.} \tag{5.54}$$

This is exact only when the M sector clusters as pressureless matter and the V sector remains smooth.

5.15 Directional extension

A minimum directional extension may be represented by a vector order parameter B_μ,

$$\boxed{\mathcal{L}_{dir} = -\frac{1}{4}Z_B(\chi)B_{\mu\nu}B^{\mu\nu} - U_B(\chi, B_\mu B^\mu),} \tag{5.55}$$

with $B_{\mu\nu} = \nabla_\mu B_\nu - \nabla_\nu B_\mu$. A nonzero cosmological expectation value selects a direction and can source dipolar or quadrupolar expansion. However, it can also generate local preferred-frame effects. The baseline fit therefore sets $B_\mu = 0$ and treats anisotropy as a nested extension whose parameters are penalized in model selection.

A genuine anisotropy prediction must specify the field equation, angular multipole, direction, amplitude, redshift dependence, observable dependence, and local preferred-frame suppression mechanism before the final data fit.

5.16 Stability conditions

The minimum action must satisfy:

$$P_{G,X} + 2X_\chi P_{G,XX} > 0, \qquad c_{s,G}^2 > 0, \qquad M_{Pl}^2 > 0, \tag{5.56}$$

$$Z_V(\chi) > 0, \qquad Z_M(\chi) > 0, \qquad Z_B(\chi) > 0, \tag{5.57}$$

and a combined effective potential bounded from below in the physical domain. Characteristic propagation cones must remain compatible with the causal structure of the effective theory.

5.17 Complete ΛCDM Correspondence Point

The minimum theory reduces to ΛCDM when

$$\boxed{n = 3, \quad Q = 0, \quad p_M = 0, \quad p_V = -\rho_V,} \tag{5.58}$$

$$\boxed{c_{s,M}^2 = 0, \quad \delta\rho_V = 0, \quad \gamma_g = 1.} \tag{5.59}$$

The Friedmann equation becomes

$$\frac{H^2}{H_0^2} = \Omega_r a^{-4} + \Omega_b a^{-3} + \Omega_c a^{-3} + \Omega_\Lambda. \tag{5.60}$$

Observable departures require at least one of $n \neq 3$, nonzero exchange, nonstandard sound speed, anisotropic stress, directional modes, strong-field deviations, or a future branch not present in ΛCDM.

5.18 Action theorems

Theorem 7 (Covariant total conservation). Diffeomorphism invariance and the field equations imply

$$\boxed{\nabla_\mu T^{\mu\nu}_{total} = 0.} \tag{5.61}$$

Theorem 8 (Exact ΛCDM background). The target equation of state and standard conservation yield exactly pressureless matter plus constant vacuum density at $n = 3$.

Theorem 9 (Exchange interpretation). If the unified component consists of pressureless M and vacuum-like V sectors with $\rho_M/\rho_V = (a_t/a)^n$, the required exchange rate is

$$Q = (3 - n)H \frac{\rho_M \rho_V}{\rho_M + \rho_V}. \tag{5.62}$$

5.19 Action-sector falsification conditions

The action requires major revision if it produces ghosts, gradient instabilities, unacceptable sound speed, violation of total conservation, composition-dependent metric coupling, an excessively varying gravitational constant, large fifth forces, an incorrect ΛCDM limit, a background fit that destroys perturbation growth, an interaction chosen after viewing validation data, forbidden preferred-frame effects, or a cyclic claim without a genuine $H = 0$ solution.

The next chapter derives the cosmological equations and observables used in the validation paper.

6 Correspondence Hierarchy, Open Extensions, and Falsifiability

6.1 The need for a correspondence hierarchy

A new framework is not established merely by reproducing familiar equations after freely choosing suitable functions. It must specify the precise limits in which established theories are recovered, the domain in which new effects appear, which quantities are derived and which remain phenomenological, and which observations can falsify the model.

The hierarchy is

$$\boxed{\text{microscopic graph and clock} \to \text{emergent quantum spacetime} \to \text{relativistic gravity} \to \text{cosmological evolution.}} \tag{6.1}$$

Each arrow is a controlled coarse-graining or limiting procedure. Failure at an earlier level invalidates the higher levels.

6.2 Levels of theoretical status

Every statement in the framework belongs to one of four categories.

6.2.1 Level I: Foundational postulates

These define the present theory, for example

$$N_{grid} = \text{constant}, \qquad K \in \mathbb{Z}, \qquad |\Psi_{K+1}\rangle = U_K|\Psi_K\rangle. \tag{6.2}$$

6.2.2 Level II: Correspondence results

These follow mathematically under specified approximations and reproduce established physics, including the Lorentz factor, Schrödinger equation, and Newtonian Poisson equation.

6.2.3 Level III: Effective model predictions

These follow from a specified effective action and a fixed parameter set, such as a derived $w_G(a)$, a sound speed, or a fixed anisotropy amplitude.

6.2.4 Level IV: Open possibilities

These are compatible with the framework but not yet derived: a regular black-hole core, a cyclic future, a nonzero preferred cosmic direction, a particular Planck-scale dispersion correction, or a microscopic derivation of the Born rule.

Open possibilities must not be presented as established predictions.

6.3 Special-relativistic correspondence

The special-relativistic limit is

$$\boxed{\frac{E}{E_P} \to 0, \qquad \frac{p}{p_P} \to 0, \qquad \chi \to 0, \qquad \nabla\chi \to 0.} \tag{6.3}$$

In this regime, the reference grid is homogeneous and isotropic, $k\ell_P \ll 1$, $\omega t_P \ll 1$, and the effective interval is Minkowskian. The Lorentz transformations and

$$E^2 = p^2c^2 + m^2c^4 \tag{6.4}$$

are recovered. Any microscopic correction $\delta_{grid}(E, p)$ must satisfy

$$\frac{\delta_{grid}}{p^2c^2+m^2c^4} \to 0. \tag{6.5}$$

A fixed integer grid is not invariant under arbitrary continuous boosts. The physical requirement is that Lorentz-breaking operators be irrelevant or strongly suppressed in the long-wavelength effective action,

$$\mathcal{L}_{eff} = \mathcal{L}_{Lorentz} + \sum_j \frac{c_j}{M_*^{d_j-4}} \mathcal{O}_j^{LV}. \tag{6.6}$$

A viable model requires the dimensionless corrections to be far below experimental limits. The smallness of ℓ_P alone does not guarantee this; suppression must follow from the microscopic dynamics or symmetry.

6.4 Quantum and classical correspondence

The quantum limit follows from the discrete unitary update when

$$\frac{t_P\|\hat{H}\|}{\hbar} \ll 1, \qquad k\ell_P \ll 1. \tag{6.7}$$

Then

$$\boxed{i\hbar\,\partial_t|\Psi\rangle = \hat{H}|\Psi\rangle,} \tag{6.8}$$

with $E = \hbar\omega$ and $p = \hbar k$.

Classical behavior requires rapid decoherence, narrow phase-space distributions, stable M-like records, and negligible interference between macroscopic histories. The condition

$$\mathcal{D} = \frac{Var(\Delta S)}{2\hbar^2} \gg 1 \tag{6.9}$$

suppresses coherence, but stable records and localized attractors are also required. This correspondence explains effective classical histories but does not derive unique individual outcomes.

6.5 General-relativistic correspondence

The baseline action reduces to general relativity when all matter couples universally to one metric, the curvature term is Einstein-Hilbert, and

$$\boxed{\gamma_g \to 1, \qquad \mu(a,k) \to 1, \qquad \eta(a,k) \to 1, \qquad c_T \to c.} \tag{6.10}$$

The weak-field metric becomes

$$ds^2 = -\left(1 + \frac{2\Phi}{c^2}\right)c^2 dt^2 + \left(1 - \frac{2\Phi}{c^2}\right)d\mathbf{x}^2, \tag{6.11}$$

and $\nabla^2 \Phi = 4\pi G\rho$. The present theory supplies a substrate interpretation of gravity, but the baseline local equations remain those of GR. A new interpretation is not, by itself, a new measurable prediction.

A distinct local-gravity prediction requires a derived effect such as finite response delay, scale-dependent gravitational coupling, additional polarizations, a modified strong-field solution, or a specific high-energy correction.

6.6 Exact ΛCDM Correspondence

The complete minimum ΛCDM correspondence point is

$$\boxed{\mathcal{P}_{\Lambda CDM} = \{n = 3, Q = 0, c_{s,M}^2 = 0, \delta_V = 0, \mu = 1, \eta = 1, A_1 = 0, A_2 = 0\}.} \tag{6.12}$$

At this point,

$$\rho_G = \rho_{c0} a^{-3} + \rho_\Lambda, \tag{6.13}$$

and both the background and the minimum growth equations are those of ΛCDM.

At $n = 3$, a_t replaces the usual cold-matter-to-vacuum density ratio,

$$a_t^3 = \frac{\Omega_{c0}}{\Omega_{\Lambda 0}}. \tag{6.14}$$

Thus, a_t is not an additional background degree of freedom relative to a ΛCDM density parameter. The genuinely additional background freedom is primarily n.

6.7 Prediction hierarchy

The predictions are divided into four tiers.

6.7.1 Tier 1: Mandatory correspondence predictions

These include local invariant c, $E = \hbar\omega$, $p = \hbar k$, $\nabla^2\Phi = 4\pi G\rho$, universal free fall, and the low-energy relativistic dispersion relation.

6.7.2 Tier 2: Minimum cosmological predictions

Given the specified $w_G(a)$, the model predicts $\rho_G(a)$, $H(z)$, and the effective exchange Q.

6.7.3 Tier 3: Action-dependent predictions

These include sound speeds, structure growth, lensing, CMB evolution, anisotropic stress, and direction dependence.

6.7.4 Tier 4: Completion-dependent predictions

These include black-hole regularization, information recovery, a bounce, Planck-scale dispersion, microscopic particle content, and the Born rule.

This hierarchy prevents completion-dependent possibilities from being confused with predictions of the minimum model.

6.8 Observational model classes

The observational program should distinguish:

- **GCC-0:** $n = 3$, isotropic, $\mu = \eta = 1$; observationally equivalent to the corresponding ΛCDM model;
- **GCC-B:** n free with the minimum clustering prescription;
- **GCC-P:** full perturbation model with action-derived sound speeds, transfer, anisotropic stress, and response functions;
- **GCC-A:** anisotropic extension with action-derived angular and redshift structure;
- **GCC-S:** strong-field completion;
- **GCC-C:** cyclic completion.

These classes must not be combined into one statistical comparison unless all additional assumptions and parameters are counted.

6.9 Identifiability

If the best-fit model satisfies

$$n = 3, \qquad A_1 = A_2 = 0, \qquad \mu = \eta = 1, \tag{6.15}$$

and all strong-field observations agree with GR, then the effective equations are observationally indistinguishable from GR plus ΛCDM. In that case, the statement that dark matter and dark energy are phases of the grid is an interpretation rather than an empirically identified mechanism.

A distinct physical theory requires at least one observable for which no parameter mapping to the standard model exists,

$$\boxed{\exists\, \mathcal{O}\colon\ \mathcal{O}_{GCC} \neq \mathcal{O}_{GR/\Lambda CDM}.} \tag{6.16}$$

6.10 Statistical falsification

For minimum chi-squared values,

$$\Delta\chi^2 = \chi^2_{\min,GCC} - \chi^2_{\min,\Lambda CDM}. \tag{6.17}$$

Model selection must use

$$\Delta AIC = \Delta\chi^2 + 2\Delta k, \tag{6.18}$$

$$\Delta BIC = \Delta\chi^2 + \Delta k \ln N. \tag{6.19}$$

If the posterior gives $n = 3$ and information criteria do not improve, current background data do not require measurable V/M exchange. This constrains the deviation claim without falsifying the exact correspondence interpretation.

The perturbative model is falsified if background-fixed parameters fail growth, CMB, power-spectrum, or lensing data. A particularly important inconsistency would be a background preference for $n \neq 3$ while perturbations require $n = 3$.

A nonzero anisotropic solution is falsified if independent data constrain its predicted amplitude to zero or yield incompatible directions. Signals that disappear after footprint correction, calibration marginalization, peculiar-velocity subtraction, or look-elsewhere correction do not validate a cosmic direction.

A particular microscopic update rule is falsified if it predicts dispersion coefficients exceeding observational bounds. This does not automatically falsify every possible discrete-substrate theory, but the broader framework faces a naturalness problem if no suppression mechanism exists.

6.11 No-retuning principle

Let a theoretical model be $\mathcal{M}(\boldsymbol{\theta}, \mathcal{F})$, where $\boldsymbol{\theta}$ is a finite parameter vector and $\mathcal{F}$ denotes the action and functional choices. After parameter determination,

$$\boxed{\boldsymbol{\theta} = \boldsymbol{\theta}_*, \qquad \mathcal{F} = \mathcal{F}_*.} \tag{6.20}$$

Validation predictions are then calculated without changing either. If the functional form is modified after a failure, the result is a new model and the data used to motivate it cannot serve as independent validation.

6.12 Minimum test sequence

The logically ordered test program is:

1. relativistic consistency and suppression of preferred-frame effects;
2. unitary quantum consistency and the Schrödinger limit;
3. weak-gravity consistency, universal free fall, lensing, and $c_T = c$;
4. background cosmology with SN Ia, BAO, and $H(z)$;
5. perturbation cosmology with growth, lensing, CMB, and matter power;
6. a directional null test;
7. strong-field predictions from the nonlinear action;
8. future turnaround and bounce only after all earlier tests.

Failure at an earlier level cannot be repaired by success in a later speculative sector.

6.13 Correspondence and distinguishability theorems

Theorem 12 (Complete correspondence hierarchy). If microscopic wavelengths and periods are long compared with ℓ_P and t_P, the reference grid is homogeneous and isotropic, evolution is unitary, matter couples universally to an Einstein metric, $n = 3$, M is pressureless and clustering, V is smooth and constant-density, and all directional amplitudes vanish, then the effective theory approaches special relativity, ordinary quantum mechanics, general relativity, and ΛCDM.

Theorem 13 (Distinguishability). If every predicted probability distribution in a domain is identical to that of the standard theory after a parameter mapping, the models are observationally degenerate in that domain. Empirical distinction requires a nondegenerate observable.

Theorem 14 (Background deviation implies exchange). In the two-sector model with $p_M = 0$, $p_V = -\rho_V$, and $\rho_M/\rho_V = (a_t/a)^n$, any $n \neq 3$ implies $Q \neq 0$ for positive H, ρ_M, and ρ_V.

6.14 Core falsification matrix

The central claims and failure conditions are:

- **Emergent Lorentz symmetry:** fails with unsuppressed preferred-frame or species-dependent effects.
- **Unitary discrete quantum evolution:** fails with loss of total probability or incorrect low-energy quantum dynamics.
- **Universal emergent gravity:** fails with composition-dependent free fall or species-dependent metrics.
- **Correct weak-field geometry:** fails with incorrect lensing, time delay, frame dragging, or gravitational-wave propagation.
- **Unified cosmology:** fails if one parameter set cannot describe both background and perturbations.
- **Nonzero exchange:** fails if $n = 3$ and no interaction signature is observed.
- **Cosmic direction:** fails if a derived nonzero amplitude is constrained to zero or the axes disagree across independent data.
- **Cyclic evolution:** fails if the derived action has no stable turnaround and bounce.

The final chapter integrates the results, limitations, and research program.

6.15 Variable effective grid number and open cosmological extensions

The baseline theory fixes the cardinality of the microscopic graph,

$$N_{\text{grid}} = |\mathcal{V}| = \text{constant}. \tag{6.21}$$

This postulate must not be confused with the number of grid degrees of freedom that are dynamically active in a coarse-grained region. We therefore reserve

$$N_{\text{eff}} = N_{\text{eff}}(K, \mathbf{N}) \tag{6.22}$$

for an effective active count. A variable N_{eff} may represent activation, deactivation, or redistribution of existing microscopic capacity without literal creation or destruction of vertices. In an isotropic coarse-graining, a minimal dimensional relation is

$$a_{\text{phys}}(K, \mathbf{N}) = a_{\text{link}}(K, \mathbf{N}) \left[\frac{N_{\text{eff}}(K, \mathbf{N})}{N_{\text{eff,0}}}\right]^{1/3}. \tag{6.23}$$

Temporal variation, $\partial_K N_{\text{eff}} \neq 0$, can alter the inferred background expansion, while spatial variation, $\nabla N_{\text{eff}} \neq 0$, can generate direction-dependent observables. A reversal in the effective activation rate could provide a kinematical ingredient for cyclic evolution. None of these consequences is derived from the microscopic update law here. They are classified as completion-dependent hypotheses and must not be used to reinterpret the observational constraints of Paper II without a new perturbation closure and a new likelihood analysis.

The notation is deliberately separated from the cosmological transition parameter n. The symbol n controls the shape of the homogeneous M--V transition; N_{grid} and N_{eff} count microscopic or effectively active degrees of freedom.

7 Discussion and Conclusions

The central construction of GCC is a hierarchy rather than a single all-purpose equation. The graph $\mathcal{G}$, the update index K, microscopic locality, and unitary closed-system evolution are foundational assumptions. Lorentz kinematics, the Schrödinger and relativistic continuum limits, and weak-field general relativity are mandatory correspondence targets. The M--V sector, the cosmological interpolation, and bounded nonlinear response are effective descriptions whose validity must be tested separately.

The framework achieves a common language for three operations that are normally introduced independently: causal propagation is finite motion on the adjacency graph; quantum dynamics is a local unitary update; and geometry is the coarse-grained constitutive response of the same substrate. This common language does not yet amount to a unique microscopic theory. In particular, the Born rule, Standard-Model field content, the nonlinear constitutive map to the metric, and the strong-field saturation law remain open.

The exact $n = 3$ ΛCDM correspondence is retained as a control point rather than advertised as a discovery. Paper II performs the observational test and finds no evidence for a homogeneous exchange away from that point. Paper III examines a distinct local strong-curvature completion without retuning the homogeneous cosmology. Direction dependence, variable N_{eff}, and cyclic evolution remain explicit future extensions.

The framework is falsified in its present form if no local update rule can produce the required Lorentz and quantum limits, if universal metric coupling fails, if the $n = 3$ implementation does not reproduce ΛCDM to numerical precision, or if a proposed completion requires independent retuning for each observable. The scientifically appropriate status is therefore a discrete foundational programme with specified correspondence limits and calculable effective sectors, not a completed theory of everything.

Appendix A: Detailed Lorentz Derivation

Flat interval and general linear transformation

The dimensionless reference interval is

$$d\Sigma^2 = -dK^2 + dN^2. \tag{A.1}$$

Let

$$K' = AK + BN, \qquad N' = CK + DN. \tag{A.2}$$

The primed origin follows $N = uK$ and satisfies $N' = 0$, giving $C = -Du$ and $N' = D(N - uK)$. Preservation of the null directions $N = \pm K$ gives $K' = D(K - uN)$. Interval invariance then requires

$$D^2(1 - u^2) = 1, \tag{A.3}$$

so $D = \gamma = (1 - u^2)^{-1/2}$. Therefore,

$$K' = \gamma(K - uN), \qquad N' = \gamma(N - uK). \tag{A.4}$$

The physical forms are the Lorentz transformations. The derivations of time dilation, length contraction, simultaneity, and velocity addition follow directly as shown in Chapter 2.

Four-momentum

The invariant particle action $S = -mc^2 \int d\tau$ gives $\mathbf{p} = \gamma m \mathbf{v}$ and $E = \gamma mc^2$. The invariant norm of $P^\mu = (E/c, \mathbf{p})$ gives

$$E^2 = p^2c^2 + m^2c^4. \tag{A.5}$$

Appendix B: Discrete Quantum Continuum Limits

Exact unitary evolution and quasi-energy

For a stationary update,

$$U = \exp\left(-\frac{it_P}{\hbar}\hat{H}_{\mathcal{G}}\right), \tag{B.1}$$

and

$$|\Psi_{K+n}\rangle = U^n|\Psi_K\rangle. \tag{B.2}$$

If $U\left|u_j\right\rangle = e^{-i\theta_j}\left|u_j\right\rangle$, then

$$E_j = \frac{\hbar}{t_P}\left(\theta_j + 2\pi m_j\right), \tag{B.3}$$

showing exact quasi-energy periodicity. The low-energy principal branch is unique when $|Et_P/\hbar| \ll 1$.

Graph Laplacian expansion

On a regular grid,

$$\left(\nabla_{\mathcal{G}}^2\psi\right)_N = \frac{\psi_{N+1} - 2\psi_N + \psi_{N-1}}{\ell_P^2}. \tag{B.4}$$

Taylor expansion gives

$$\nabla_{\mathcal{G}}^2\psi = \partial_x^2\psi + \frac{\ell_P^2}{12}\partial_x^4\psi + \mathcal{O}(\ell_P^4). \tag{B.5}$$

The lattice kinetic energy is

$$E_{\mathcal{G}}(k) = \frac{2\hbar^2}{m\ell_P^2}\sin^2\left(\frac{k\ell_P}{2}\right) = \frac{\hbar^2k^2}{2m} - \frac{\hbar^2k^4\ell_P^2}{24m} + \cdots. \tag{B.6}$$

Gaussian decoherence

For a Gaussian relative-action fluctuation,

$$\Gamma = \left\langle e^{i\Delta S/\hbar}\right\rangle = \exp\left[\frac{i\langle\Delta S\rangle}{\hbar} - \frac{Var(\Delta S)}{2\hbar^2}\right]. \tag{B.7}$$

Thus

$$|\Gamma| = e^{-\mathcal{D}}, \qquad \mathcal{D} = \frac{Var(\Delta S)}{2\hbar^2}. \tag{B.8}$$

Appendix C: Variation of the Effective Action

For

$$S_G = \int d^4x\sqrt{-g}\,P_G(\chi, X_\chi), \tag{C.1}$$

with $X_\chi = -g^{\mu\nu}\nabla_\mu\chi\nabla_\nu\chi/2$, metric variation gives

$$T^{(G)}_{\mu\nu} = P_{G,X}\nabla_\mu\chi\nabla_\nu\chi + P_G g_{\mu\nu}. \tag{C.2}$$

For a timelike field gradient,

$$p_G = P_G, \qquad \rho_G = 2X_\chi P_{G,X} - P_G. \tag{C.3}$$

Scalar variation gives

$$\nabla_\mu\left(P_{G,X}\nabla^\mu\chi\right) + P_{G,\chi} = \mathcal{S}_\chi^{int}. \tag{C.4}$$

The metric equation is

$$M_{Pl}^2 G_{\mu\nu} = T_{\mu\nu}^{total}. \tag{C.5}$$

The contracted Bianchi identity implies

$$\nabla_\mu T^{\mu\nu}_{total} = 0. \tag{C.6}$$

The scalar kinetic coefficient and sound speed are

$$\mathcal{K}_\chi = P_{G,X} + 2X_\chi P_{G,XX}, \tag{C.7}$$

$$c_{s,G}^2 = \frac{P_{G,X}}{P_{G,X} + 2X_\chi P_{G,XX}}. \tag{C.8}$$

Ghost and gradient stability require $\mathcal{K}_\chi > 0$ and $c_{s,G}^2 > 0$.

Appendix E: Notation and Dimensions

The manuscript uses metric signature $(-,+,+,+)$. The scale factor is dimensionless with $a_0 = 1$, and $1 + z = a^{-1}$. A dot denotes d/dt and a prime denotes $d/d\ln a$ unless stated otherwise.

Fundamental substrate symbols

$\mathcal{G} = (\mathcal{V}, \mathcal{E})$ -- Fundamental grid graph; dimensionless relational structure.

N_{grid} -- Total number of elementary grid units; dimensionless.

K -- Universal integer update index; dimensionless.

N^i -- Grid-coordinate count in spatial direction i; dimensionless.

$\mathcal{X}_K$ -- Complete microscopic state at update K.

U_K -- Unitary update operator; dimensionless.

t_P, ℓ_P -- Planck-time and Planck-length calibration scales; time and length.

E_P, p_P -- Planck energy and Planck momentum; energy and momentum.

Emergent spacetime and gravity symbols

α -- Emergent lapse or local clock-rate factor; dimensionless.

β^i -- Emergent shift or directional grid flow; dimensionless.

h_{ij} -- Emergent spatial metric; dimensionless in the adopted coordinate convention.

$g_{\mu\nu}$ -- Effective spacetime metric.

τ, t, ℓ -- Proper time, cosmological coordinate time, and physical distance.

$a(t)$ -- Cosmological scale factor; dimensionless.

χ -- Scalar grid-state or processing-load variable; dimensionless.

J^i -- Directional grid-state current; normalized and dimensionless.

Π^{ij} -- Anisotropic grid-state tensor; normalized and dimensionless.

Φ, Ψ -- Newtonian and spatial-curvature metric potentials; dimensions of velocity squared.

γ_g -- Weak-field ratio of spatial to temporal metric response; dimensionless.

$\mu(a,k)$, $\eta(a,k)$, $\Sigma(a,k)$ -- Effective Newtonian, gravitational-slip, and lensing-response functions; dimensionless.

Quantum and V/M Symbols

$|\Psi_K\rangle$ -- Quantum state at update K.

$\psi_i(K)$ -- Grid-basis probability amplitude; dimensionless after normalization.

$\hat{H}$ -- Hamiltonian generator; energy.

S -- Action; energy multiplied by time.

$\hbar$, h -- Reduced and ordinary Planck constants; action.

ω, ν, k, p -- Angular frequency, frequency, wave number, and momentum.

V, M -- Coherent extended and localized stable effective sectors.

P_V, P_M -- Effective projectors onto the V and M sectors.

N_{eff} -- Effective participation number; dimensionless.

C_{off} -- Off-diagonal coherence measure; dimensionless.

Γ -- Coherence factor; dimensionless.

$\mathcal{D}$ -- Decoherence exponent; dimensionless.

L_V, L_M -- Coherence and localization length scales.

Effective-action and cosmological symbols

$P_G(\chi, X_\chi)$ -- Grid-field Lagrangian density; energy density.

X_χ -- Grid-field kinetic invariant; inverse length squared for dimensionless χ.

$Z_G(\chi)$, $U_G(\chi)$ -- Grid kinetic-response function and grid-state potential.

Q_A^μ, Q -- Covariant sector-transfer vector and homogeneous energy-transfer rate.

$c_{s,G}^2$ -- Grid-field rest-frame sound speed squared in units of c^2; dimensionless.

H, H_0 -- Hubble expansion rate and its present value; inverse time.

$E(a) = H(a)/H_0$ -- Normalized Hubble function; dimensionless.

a_t, n -- Transition scale factor and effective V/M exchange parameter; dimensionless.

w_G, F_G -- Grid equation-of-state parameter and normalized density; dimensionless.

$\rho_r, \rho_b, \rho_G, \rho_M, \rho_V$ -- Radiation, baryon, total-grid, matter-like, and vacuum-like energy densities.

Ω_{A0} -- Present density fraction of component A; dimensionless.

D_+, f, σ_8 -- Linear growth factor, logarithmic growth rate, and fluctuation amplitude; dimensionless.

$D_C, D_M, D_L, D_A, D_H, D_V$ -- Standard cosmological distance measures; length.

A_1, A_2 -- Dipolar and quadrupolar directional amplitudes; dimensionless.

$\hat{\mathbf{n}}$, $\hat{\mathbf{p}}$ -- Line-of-sight and preferred-direction unit vectors.

Index and derivative conventions

Greek indices μ, ν, α, β run over spacetime coordinates. Latin indices i, j, k run over spatial coordinates. Capital Latin indices may denote dimensionless grid-clock spacetime coordinates. Repeated indices are summed unless stated otherwise.

The covariant derivative is ∇_μ, the local spatial Laplacian is ∇^2, and the graph Laplacian is $\nabla_{\mathcal{G}}^2$.

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